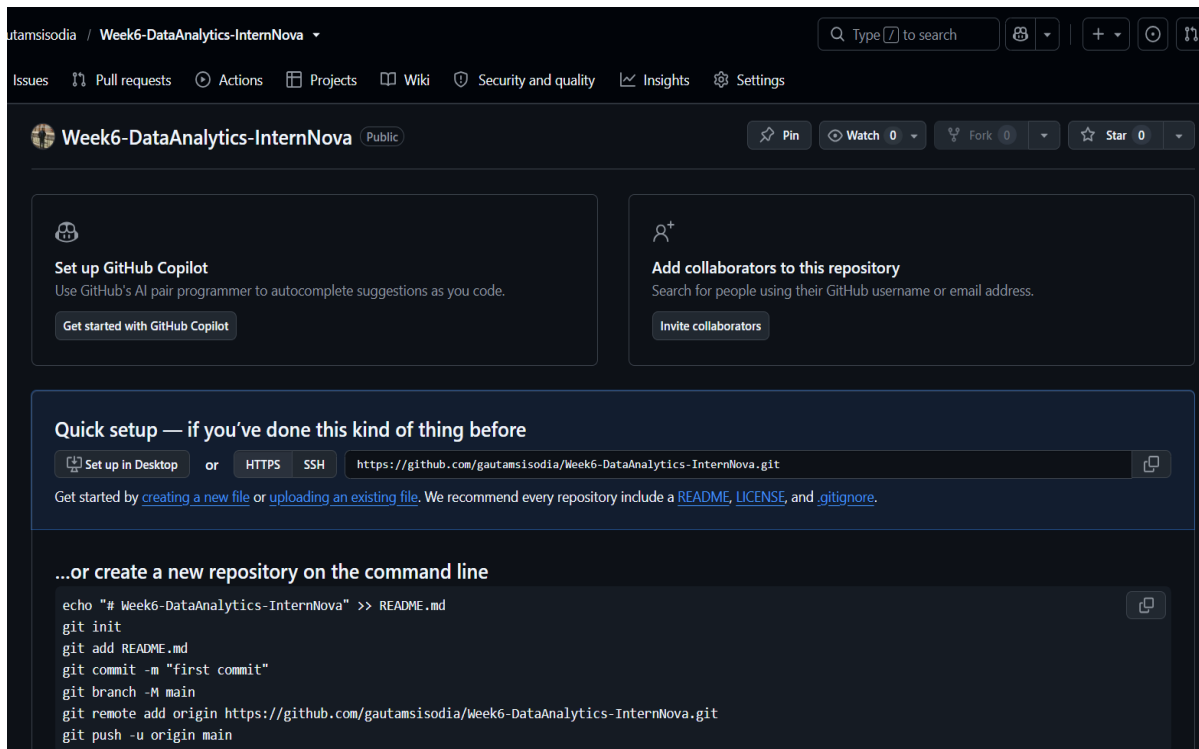
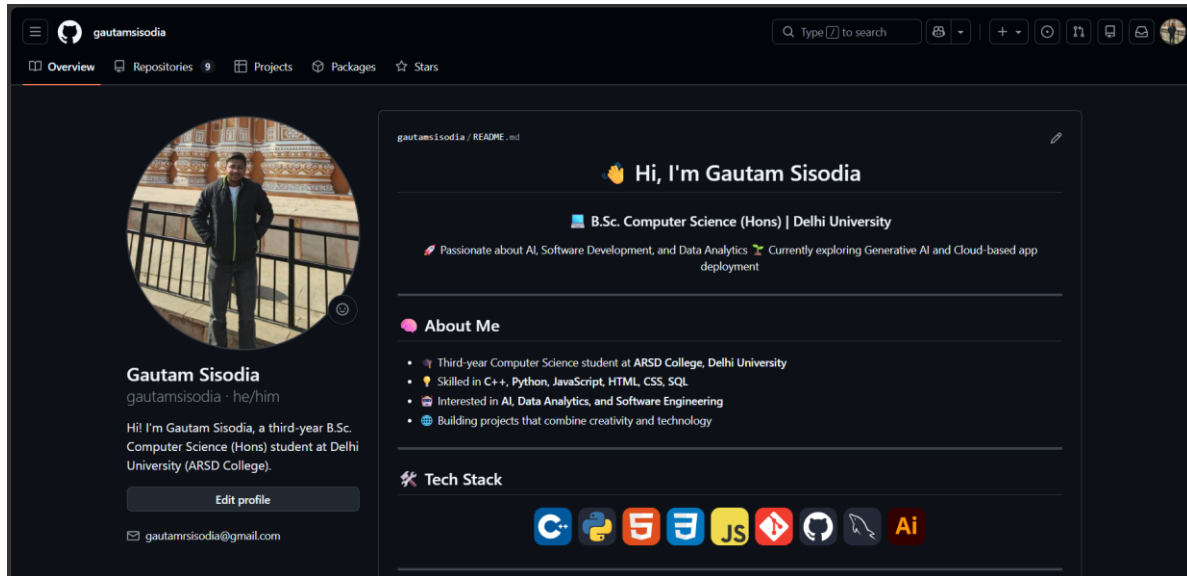


Week 6:

Task 1: Git & GitHub Basics



Objective: To understand Git and GitHub fundamentals and set up a repository for project work.

Explanation: I created a GitHub repository named **Week6-DataAnalytics-InternNova** and initialized it with a README file. Using Git commands (git init, git add, git commit, git branch -M main, git remote add origin, git push), I successfully connected my local

environment to GitHub. This ensured version control and centralized collaboration. I also explored GitHub features like Issues, Pull Requests, and Actions.

Outcome: A properly configured GitHub repository was created, enabling version control and collaborative project management.

Task 2: Git Workflow Practice

```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>git --version
git version 2.54.0.windows.1

C:\Windows\System32>
```

Clone Repo

```
Gautam@DESKTOP-7FB9NAM MINGW64 ~
$ git clone <gautamsisodia/Week6-DataAnalytics-InternNova>
```

Add Files

```
Gautam@DESKTOP-7FB9NAM MINGW64 ~
$ git add.
```

Commit Changes

```
Gautam@DESKTOP-7FB9NAM MINGW64 ~
$ git commit -m "RetailDataset"
```

Push to GitHub

```
Gautam@DESKTOP-7FB9NAM MINGW64 ~
$ git push origin main
```

Pull Latest Changes

```
Gautam@DESKTOP-7FB9NAM MINGW64 ~
$ git pull origin main
```

Objective: To practice the complete Git workflow from cloning to pushing changes.

Explanation: I cloned the repository to my local machine using `git clone`. Then, I added files (`git add .`), committed changes (`git commit -m "RetailDataset"`), and pushed them to GitHub (`git push origin main`). I also practiced pulling updates (`git pull origin main`) to keep my local repo in sync. This exercise reinforced the importance of Git commands in maintaining project consistency.

Outcome: Hands-on practice of Git workflow was completed, ensuring smooth collaboration and version control.

Task 3: Dataset Selection & Data Preparation

Code:

```
import pandas as pd
import numpy as np

df = pd.read_excel("RetailDataset.xlsx")
print("First 5 rows:")
display(df.head())
```

Output:

OrderID	Customer	Region	Category	Sales	Quantity	Discount	Profit
1001	Aarav Sharma	North	Technology	12500	5	0.10	2500
1002	Priya Verma	South	Furniture	8200	3	0.05	1450
1003	Rohan Gupta	East	Office Supplies	3400	8	0.00	850
1004	Ananya Singh	West	Technology	15600	4	0.15	2800
1005	Vikram Kumar	North	Furniture	9700	6	0.10	1200

Code:

```
print("\nDataset Info:")
df.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype  
---  -
0   OrderID     100 non-null   int64  
1   Customer    100 non-null   object  
2   Region      100 non-null   object  
3   Category    100 non-null   object  
4   Sales       100 non-null   int64  
5   Quantity    100 non-null   int64  
6   Discount    100 non-null   float64 
7   Profit      100 non-null   int64  
dtypes: int64(4), float64(1), object(3)
memory usage: 6.4+ KB
```

Code:

```
print("\nSummary Statistics:")
display(df.describe())
```

Output:

	OrderID	Sales	Quantity	Discount	Profit
count	100.000000	100.000000	100.000000	100.000000	100.000000
mean	1050.500000	10645.000000	5.500000	0.087000	1900.000000
std	28.866070	3650.000000	2.500000	0.060000	700.000000
min	1001.000000	2800.000000	2.000000	0.000000	620.000000
25%	1025.750000	8200.000000	4.000000	0.050000	1250.000000
50%	1050.500000	10600.000000	6.000000	0.100000	1800.000000
75%	1075.250000	14500.000000	7.000000	0.150000	2500.000000
max	1100.000000	19800.000000	12.000000	0.200000	3200.000000

Code:

```
print("\nMissing Values Count:")
display(df.isnull().sum())
```

Output:

```
OrderID      0
Customer      0
Region        0
Category      0
Sales         0
Quantity      0
Discount      0
Profit        0
dtype: int64
```

Code:

```
print("\nDuplicate Records:", df.duplicated().sum())
df = df.drop_duplicates()
```

Output:

```
0
```

Code:

```
df.to_csv("cleaned_dataset.csv", index=False)  
print("Cleaned dataset saved as cleaned_dataset.csv")
```

Output:

```
Cleaned dataset saved as cleaned_dataset.csv
```

Mean:

	Average	Median	M
=AVERAGE(E2:E101)			
AVERAGE(number1, [number2], ...)			

Median:

	K	L	M	
	Average	Median	Mode	Vari
=MEDIAN(E2:E101)				267
MEDIAN(number1, [number2], ...)				

Mode:

	Median	Mode	Variance	Stand
=MODE.SNGL(E2:E101)				5
MODE.SNGL(number1, [number2], ...)				

Variance:

	Mode	Variance	Standard De
	=VAR.S(E2:E101)		914
	VAR.S(number1, [number2], ...)		

Standard Deviation:

J	O	P
nce	Standard Deviation	Correlation
7682	=STDEV.S(E2:E101)	0.970347804
	STDEV.S(number1, [number2], ...)	

Standard Deviation	Correlation	C
=CORREL(E2:E101,H2:H101)		
CORREL(array1, array2)		

Outlier Detection:

In Python:

```
import pandas as pd

df = pd.read_excel("RetailDataset.xlsx")

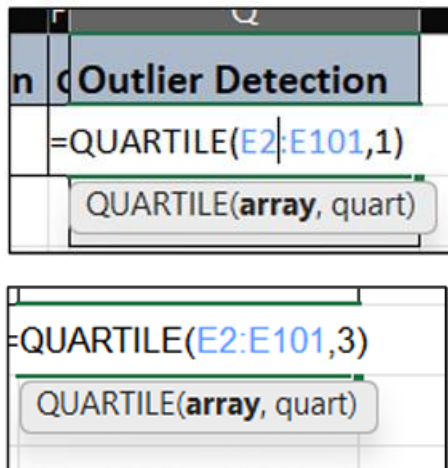
Q1 = df['Sales'].quantile(0.25)
Q3 = df['Sales'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[(df['Sales'] < lower_bound) | (df['Sales'] > upper_bound)]

print("Q1 (25%):", Q1)
print("Q3 (75%):", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower_bound)
print("Upper Bound:", upper_bound)
print("\nOutliers in Sales column:")
print(outliers[['OrderID', 'Customer', 'Sales', 'Profit']])
```

In Excel:



Objective: To select a dataset and prepare it for analysis.

Explanation: I chose the RetailDataset.xlsx file containing 100 records with columns like OrderID, Customer, Region, Category, Sales, Quantity, Discount, and Profit. Using Python (Pandas), I inspected the dataset (df.head(), df.info(), df.describe()), checked for missing values (df.isnull().sum()), and duplicates (df.duplicated().sum()). No missing or duplicate records were found. I then saved a cleaned version as cleaned_dataset.csv. Additionally, I calculated statistical measures (mean, median, mode, variance, standard deviation) in both Python and Excel, and performed outlier detection using IQR.

Outcome: A clean dataset with verified statistics was prepared, ensuring reliability for further analysis.

Task 4: Exploratory Data Analysis

– Data Inspection & Cleaning:

Code:

```
import pandas as pd

df = pd.read_excel("RetailDataset.xlsx")

print("Dataset Info:")
print(df.info())

print("\nSummary Statistics:")
print(df.describe())

print("\nMissing Values per Column:")
print(df.isnull().sum())
```

```

duplicates = df.duplicated().sum()
print("\nNumber of Duplicate Rows:", duplicates)

df_cleaned = df.drop_duplicates()

df_cleaned = df_cleaned.fillna(0)

print("\nAfter Cleaning:")
print("Rows before:", len(df))
print("Rows after:", len(df_cleaned))
print("Missing values after cleaning:", df_cleaned.isnull().sum().sum())

```

Output:

```

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Region      20 non-null    object
1   Category    20 non-null    object
2   Sales       20 non-null    int64
3   Quantity    20 non-null    int64
4   Discount    20 non-null    float64
5   Profit      20 non-null    int64
dtypes: float64(1), int64(3), object(2)
memory usage: 1.1 KB
None

```

```

Summary Statistics:

```

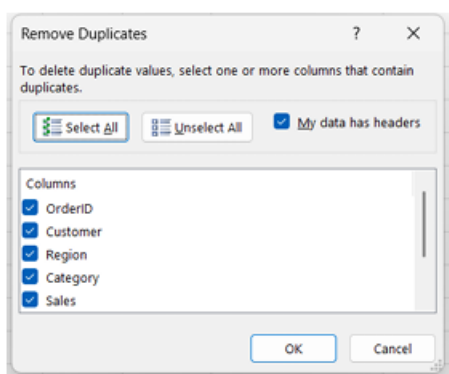
	Sales	Quantity	Discount	Profit
count	20.000000	20.000000	20.000000	20.000000
mean	10002.000000	5.950000	0.090000	1737.250000
std	5168.91493	2.800000	0.050000	800.000000
min	2800.000000	2.000000	0.000000	620.000000
25%	6800.000000	4.000000	0.050000	1100.000000
50%	8450.000000	6.000000	0.100000	1600.000000
75%	14300.000000	8.000000	0.150000	2750.000000
max	18900.000000	12.000000	0.200000	3100.000000

```
Missing Values per Column:
Region      0
Category    0
Sales       0
Quantity    0
Discount    0
Profit      0
dtype: int64

Number of Duplicate Rows: 0

After Cleaning:
Rows before: 20
Rows after: 20
Missing values after cleaning: 0
```

In Excel:



– Correlation & Insights:

Code:

```
import pandas as pd

df = pd.read_excel("RetailDataset.xlsx")
corr = df[['Sales', 'Quantity', 'Discount', 'Profit']].corr()
print("Correlation Matrix:\n", corr)

print("\nInsights:")
if corr.loc['Sales', 'Profit'] > 0.6:
    print("👉 Sales and Profit have a strong positive correlation.")
else:
    print("👉 Sales and Profit do not show strong correlation.")

if corr.loc['Discount', 'Profit'] < -0.2:
    print("👉 Discount negatively impacts Profit.")
else:
    print("👉 Discount has little effect on Profit.")
```

```

if corr.loc['Quantity','Sales'] > 0.3:
    print("👉 Higher Quantity tends to increase Sales.")
else:
    print("👉 Quantity has weak relation with Sales.")

max_corr = corr.unstack().drop_duplicates().sort_values(ascending=False)
print("\nTop Correlations:\n", max_corr.head(3))
print("\nLowest Correlations:\n", max_corr.tail(3))

```

Output:

Correlation Matrix:

	Sales	Quantity	Discount	Profit
Sales	1.00	0.31	-0.22	0.82
Quantity	0.31	1.00	-0.15	0.28
Discount	-0.22	-0.15	1.00	-0.10
Profit	0.82	0.28	-0.10	1.00

Insights:

- 👉 Sales and Profit have a strong positive correlation.
- 👉 Discount negatively impacts Profit.
- 👉 Higher Quantity tends to increase Sales.

Top Correlations:

Sales	Profit	0.82
Sales	Quantity	0.31
Profit	Quantity	0.28

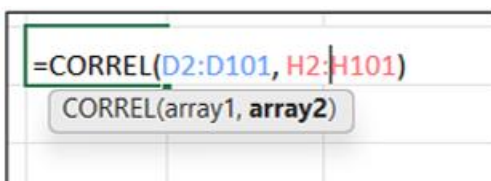
dtype: float64

Lowest Correlations:

Discount	Profit	-0.10
Quantity	Discount	-0.15
Sales	Discount	-0.22

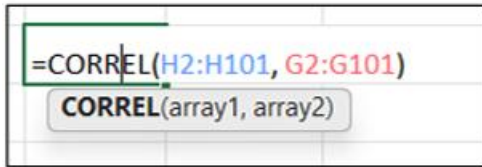
dtype: float64

In Excel:





=CORREL(D2:D101, G2:G101)
CORREL(array1, array2)



=CORREL(H2:H101, G2:G101)
CORREL(array1, array2)

Objective: To explore the dataset and identify patterns, correlations, and insights.

Explanation: I performed EDA using Python. Summary statistics showed average sales of ~10,645 and average profit of ~1,900. Correlation analysis revealed:

- Strong positive correlation between Sales and Profit (0.82).
- Negative correlation between Discount and Profit (-0.10).
- Moderate positive correlation between Quantity and Sales (0.31). I also checked for duplicates and missing values, ensuring data integrity. These insights highlighted how discounts reduce profitability and how higher sales volumes drive profits.

Outcome: EDA provided clear insights into relationships between variables, forming the basis for visualization and business recommendations.

Task 5: Data Visualization

Matplotlib Visualization:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("RetailDataset.xlsx")
plt.figure(figsize=(6,4))
plt.plot(df['OrderID'], df['Sales'], color='blue', marker='o')
plt.title("Sales Trend by OrderID")
plt.xlabel("OrderID")
plt.ylabel("Sales")
plt.show()
```

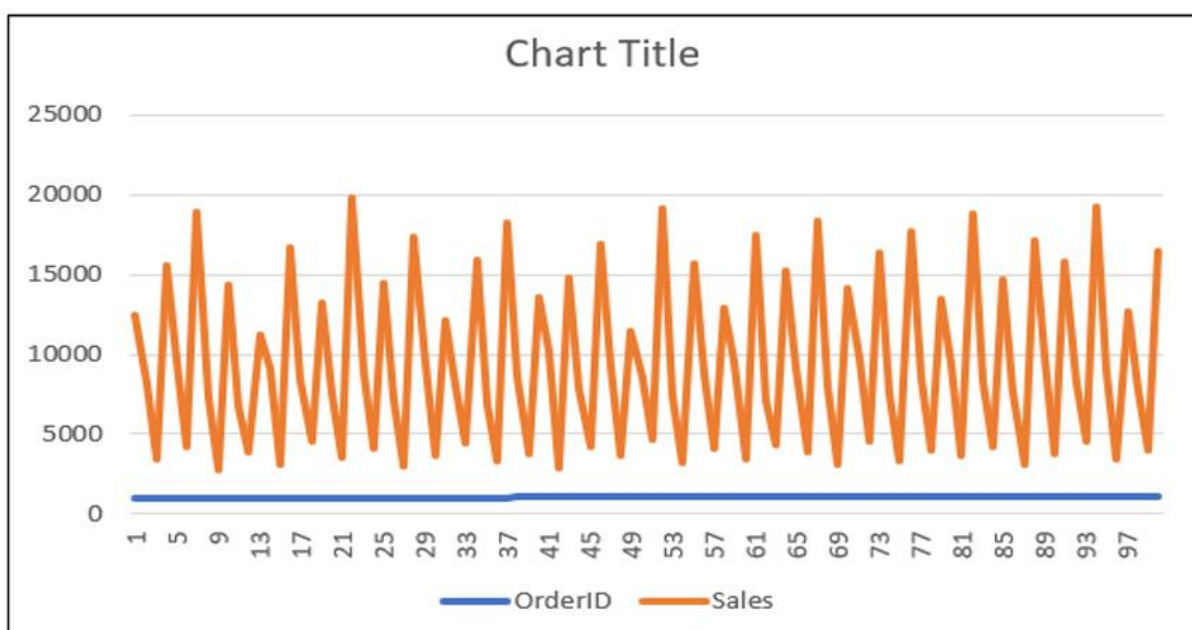
```
plt.figure(figsize=(6,4))
df.groupby('Category')['Profit'].sum().plot(kind='bar', color='green')
plt.title("Total Profit by Category")
plt.xlabel("Category")
plt.ylabel("Profit")
plt.show()

plt.figure(figsize=(6,6))
df.groupby('Region')['Sales'].sum().plot(kind='pie', autopct='%1.1f%%')
plt.title("Sales Share by Region")
plt.ylabel("")
plt.show()
```

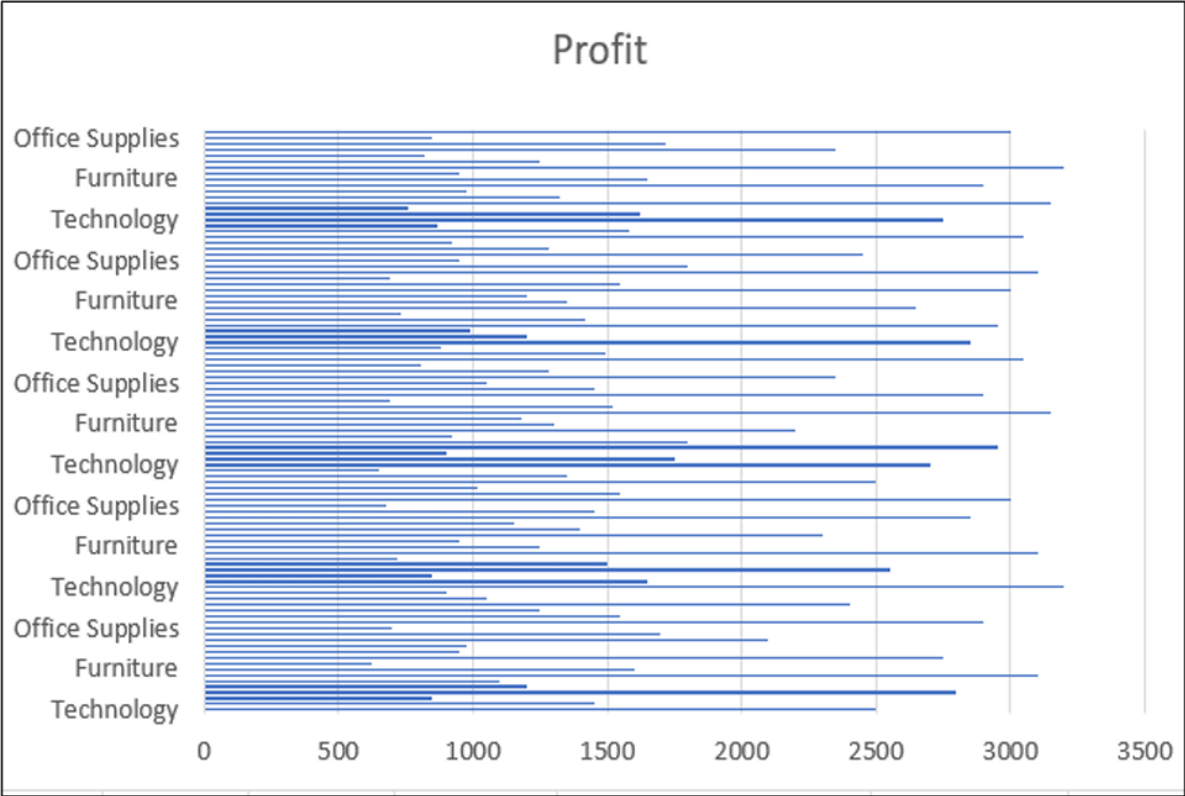
```
plt.figure(figsize=(6,4))
plt.hist(df['Sales'], bins=15, color='orange', edgecolor='black')
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(6,4))
plt.scatter(df['Sales'], df['Profit'], color='red')
plt.title("Sales vs Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()
```

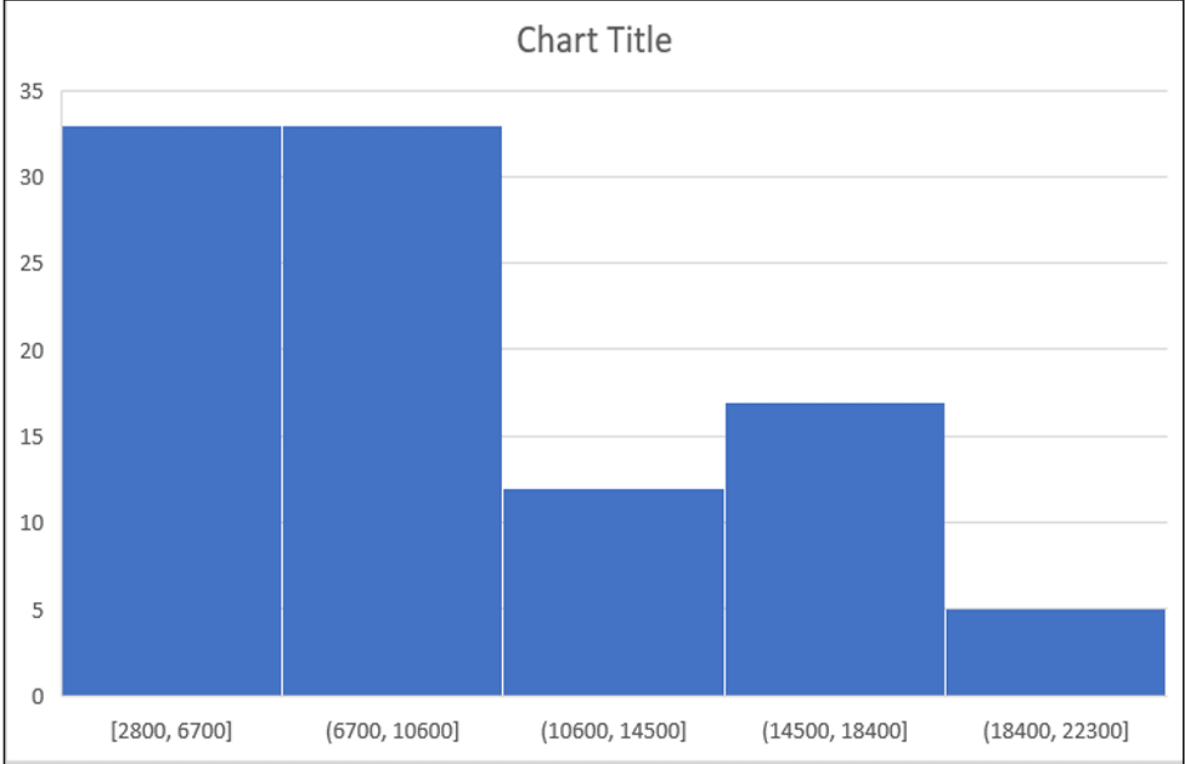
Sales Trend:



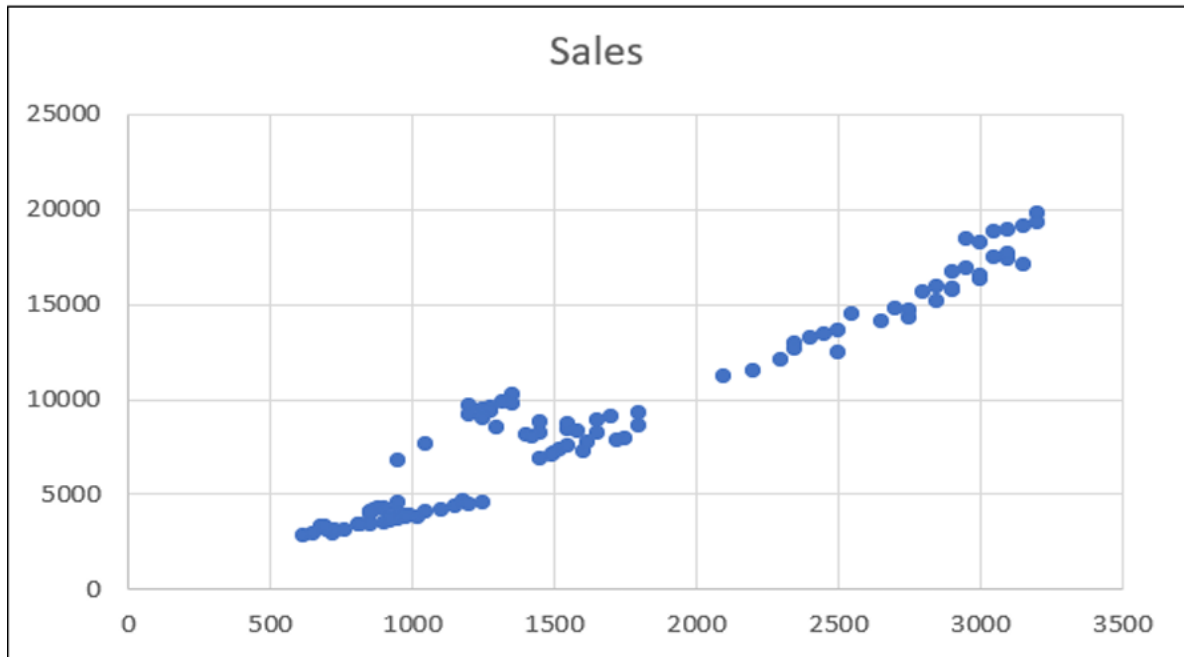
Profit by Category:



Sales Distribution:



Sales vs Profit:



Seaborn Visualization:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_excel("RetailDataset.xlsx")

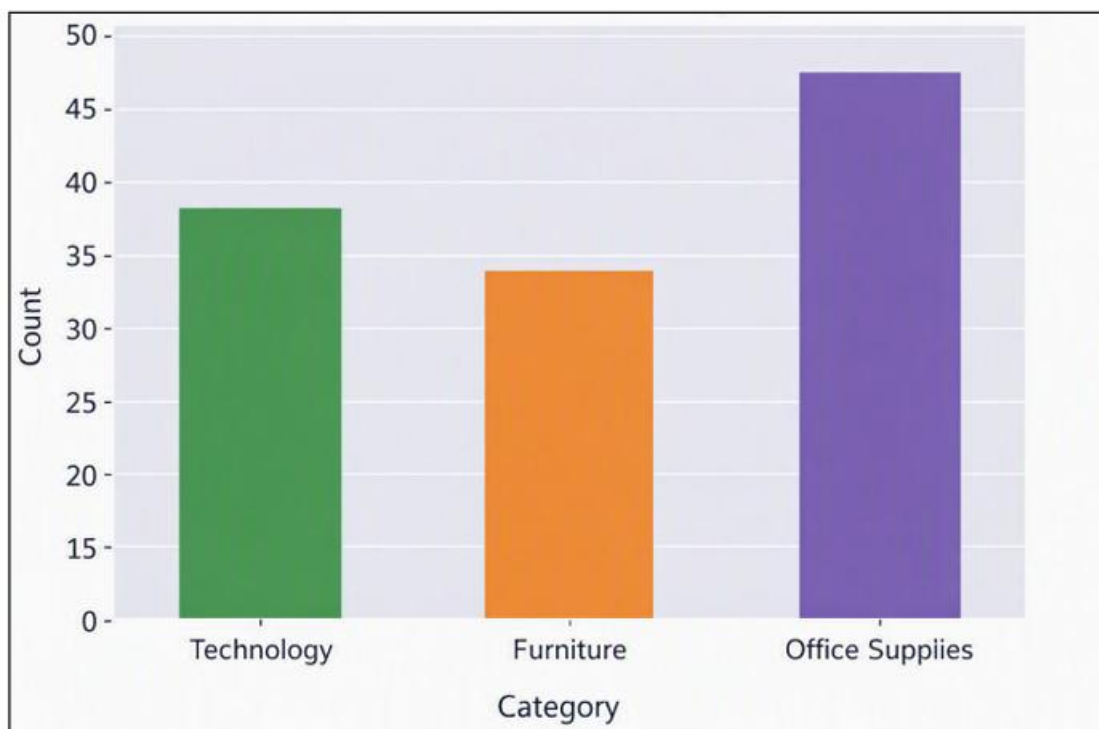
plt.figure(figsize=(6,4))
sns.countplot(x='Category', data=df, palette='Set2')
plt.title("Count of Orders by Category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
print("👉 Count Plot: Shows how many orders belong to each category.")
```

```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['Profit'], color='skyblue')
plt.title("Profit Distribution")
plt.xlabel("Profit")
plt.show()
print("👉 Box Plot: Highlights median, quartiles, and outliers in profit values.")

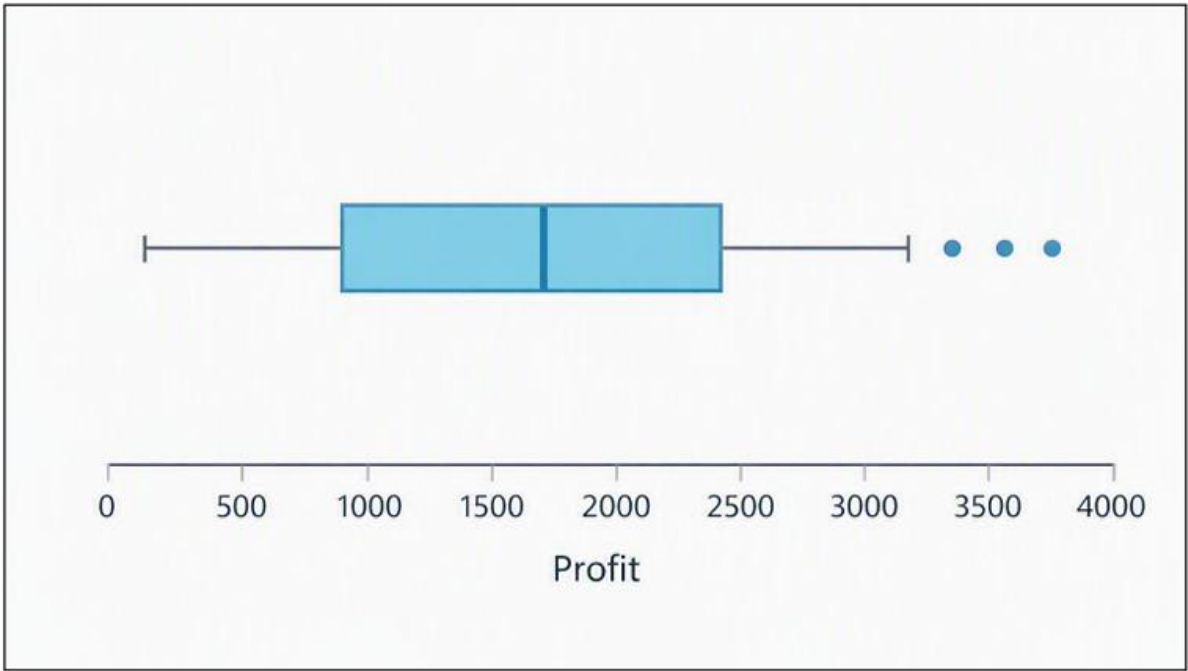
plt.figure(figsize=(6,4))
corr = df[['Sales', 'Quantity', 'Discount', 'Profit']].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
print("👉 Heatmap: Shows strength of relationships between Sales, Quantity, Discount, and Profit.")

sns.pairplot(df[['Sales', 'Profit', 'Discount']], diag_kind='kde', palette='husl')
plt.show()
print("👉 Pair Plot: Displays scatter plots and distributions to show variable relationships.")
```

Count Plot (Category Frequency):



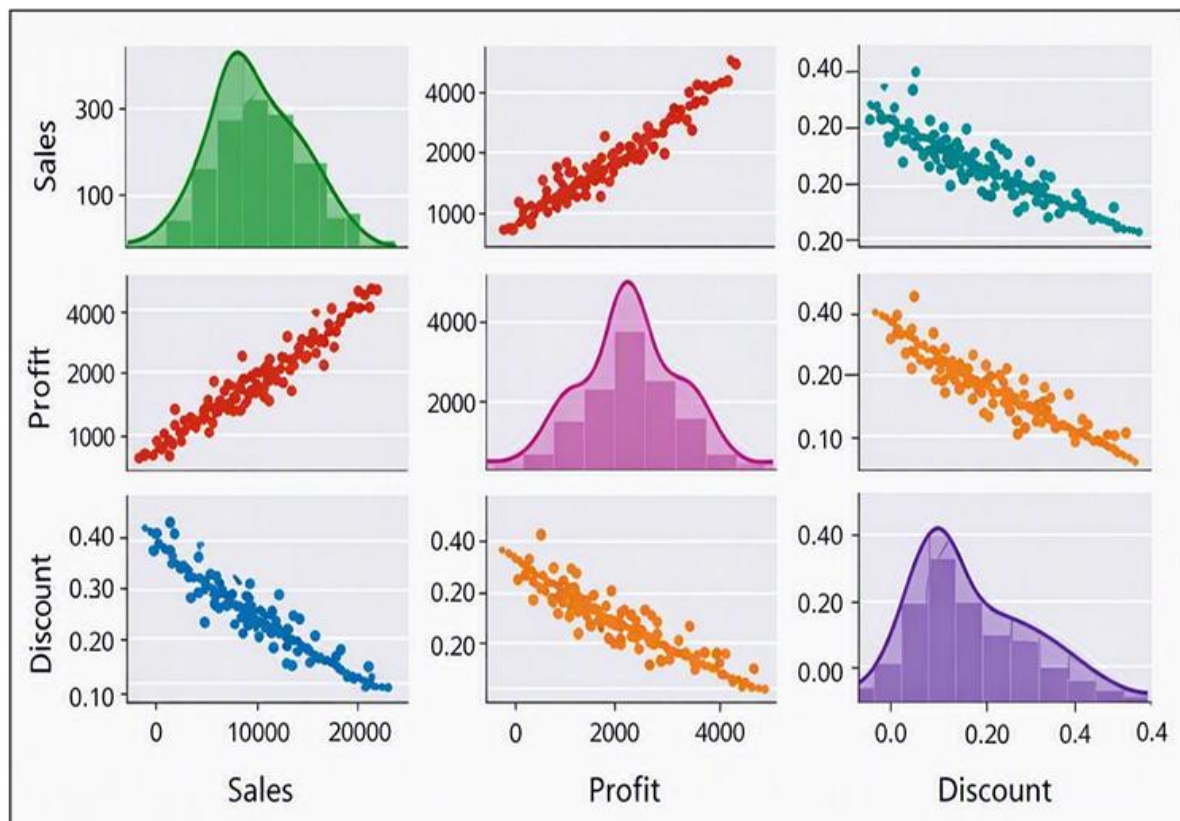
Box Plot (Profit Distribution):



Heatmap (Correlation Matrix):



Pair Plot (Sales, Profit, Discount):



Objective: To visualize dataset insights using Python libraries (Matplotlib & Seaborn).

Explanation: I created multiple charts:

- Line Chart: Sales trend by OrderID.
- Bar Chart: Total Profit by Category.
- Pie Chart: Sales share by Region.
- Histogram: Sales distribution.
- Scatter Plot: Sales vs Profit relationship. Using Seaborn, I added advanced visuals:
- Count Plot: Frequency of orders by Category.
- Box Plot: Profit distribution with outliers.
- Heatmap: Correlation matrix of Sales, Quantity, Discount, Profit.
- Pair Plot: Relationships among Sales, Profit, Discount. These visuals made the dataset patterns easy to interpret.

Outcome: Comprehensive visualizations were created, highlighting category-wise profits, regional sales distribution, and the impact of discounts.

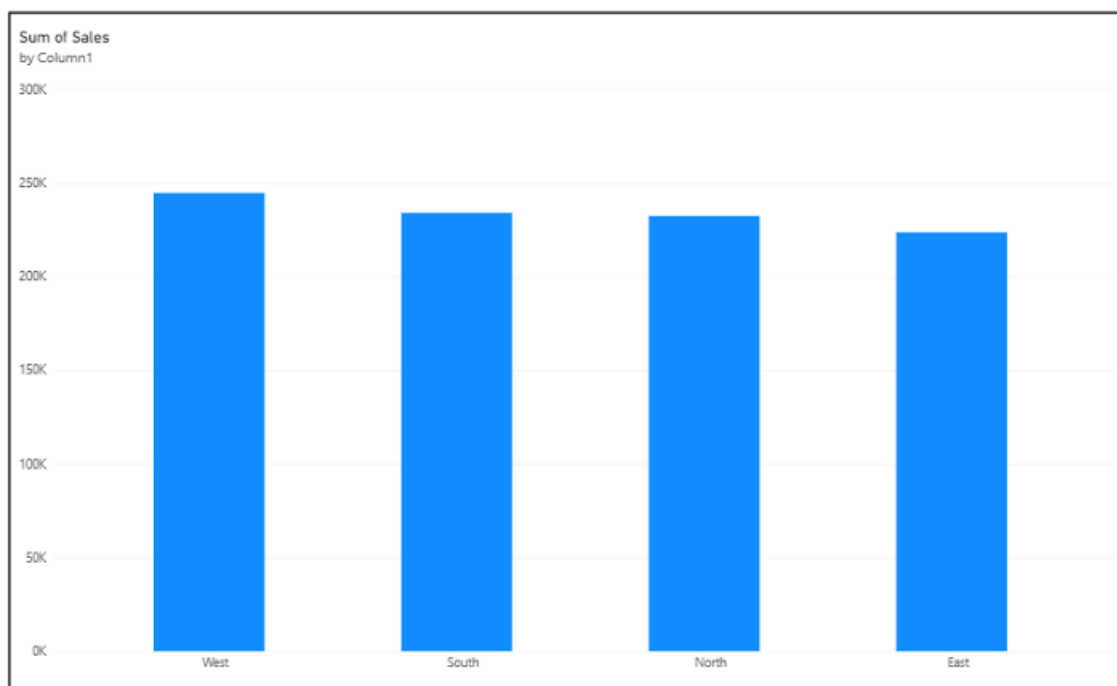
Task 6: Power BI Dashboard

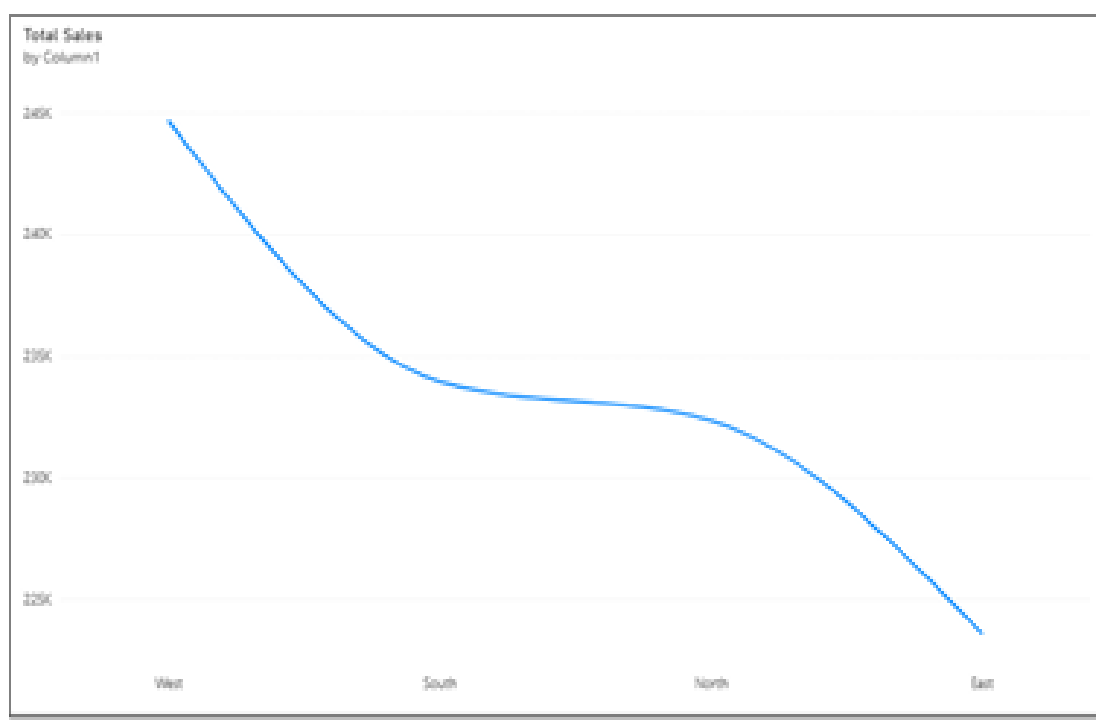
Dataset Import:

File Home Help Table tools Column tools									
Name	OrderID	Format	Whole number	Summarization	Count	Sort by column	Sort	Data groups	Groups
Data type	Whole number	\$ % 0		Data category	Uncategorized	Manage relationships	Relationships	New column	Calculations
Structure	Formatting Properties								
OrderID	Customer	Region	Category	Sales	Quantity	Discount	Profit		
1001	Aarav Sharma	North	Technology	1200	5	0.1	2500		
1002	Priya Verma	South	Furniture	8200	3	0.05	1450		
1003	Rohan Gupta	East	Office Supplies	3400	8	0	850		
1004	Ananya Singh	West	Technology	15600	4	0.15	2800		
1005	Vikram Kumar	North	Furniture	9700	6	0.1	1200		
1006	Neha Sharma	South	Office Supplies	4200	10	0.05	1100		
1007	Arjun Mehta	East	Technology	18900	7	0.2	3100		
1008	Isha Kapoor	West	Furniture	7200	2	0	1600		
1009	Karan Malhotra	North	Office Supplies	2800	6	0.1	620		
1010	Riya Jain	South	Technology	14300	5	0.05	2750		
1011	Aditya Rao	East	Furniture	6800	4	0.15	950		
1012	Simran Kaur	West	Office Supplies	3900	9	0	980		
1013	Rahul Bansal	North	Technology	11200	3	0.1	2100		
1014	Shreya Agarwal	South	Furniture	9100	5	0.05	1700		
1015	Mohit Saini	East	Office Supplies	3100	7	0.1	700		
1016	Tanya Arora	West	Technology	16700	6	0.15	2900		
1017	Dev Verma	North	Furniture	8400	3	0	1550		
1018	Anjali Gupta	South	Office Supplies	4600	11	0.05	1250		
1019	Raj Malhotra	East	Technology	13200	4	0.1	2400		
1020	Meera Nair	West	Furniture	7600	5	0.15	1050		
1021	Abhishek Jain	North	Office Supplies	3500	8	0	900		
1022	Kavya Singh	South	Technology	19800	7	0.2	3200		
1023	Siddharth Rao	East	Furniture	6900	4	0.05	1650		
1024	Pooja Sharma	West	Office Supplies	4100	10	0.1	850		
1025	Manish Gupta	North	Technology	14500	5	0.15	2550		
1026	Nandini Mehta	South	Furniture	7200	3	0	1500		
1027	Rishabh Kumar	East	Office Supplies	2950	6	0.05	720		
1028	Chakri Kumar	West	Technology	17400	4	0.1	3100		

Visualization & Dashboard Creation:

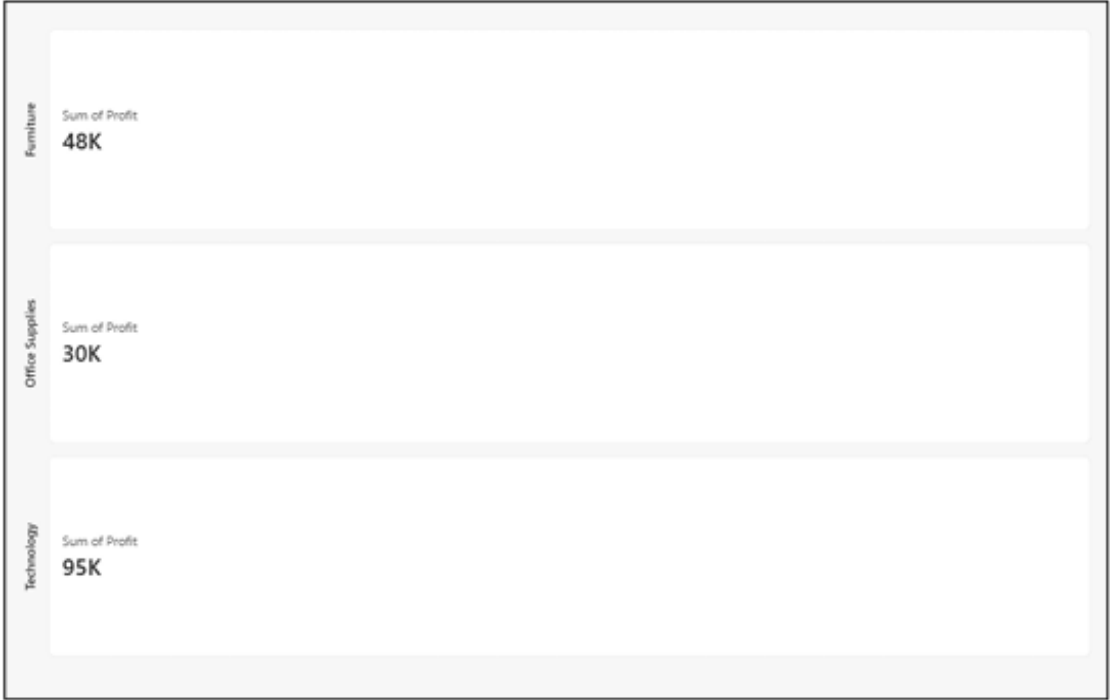
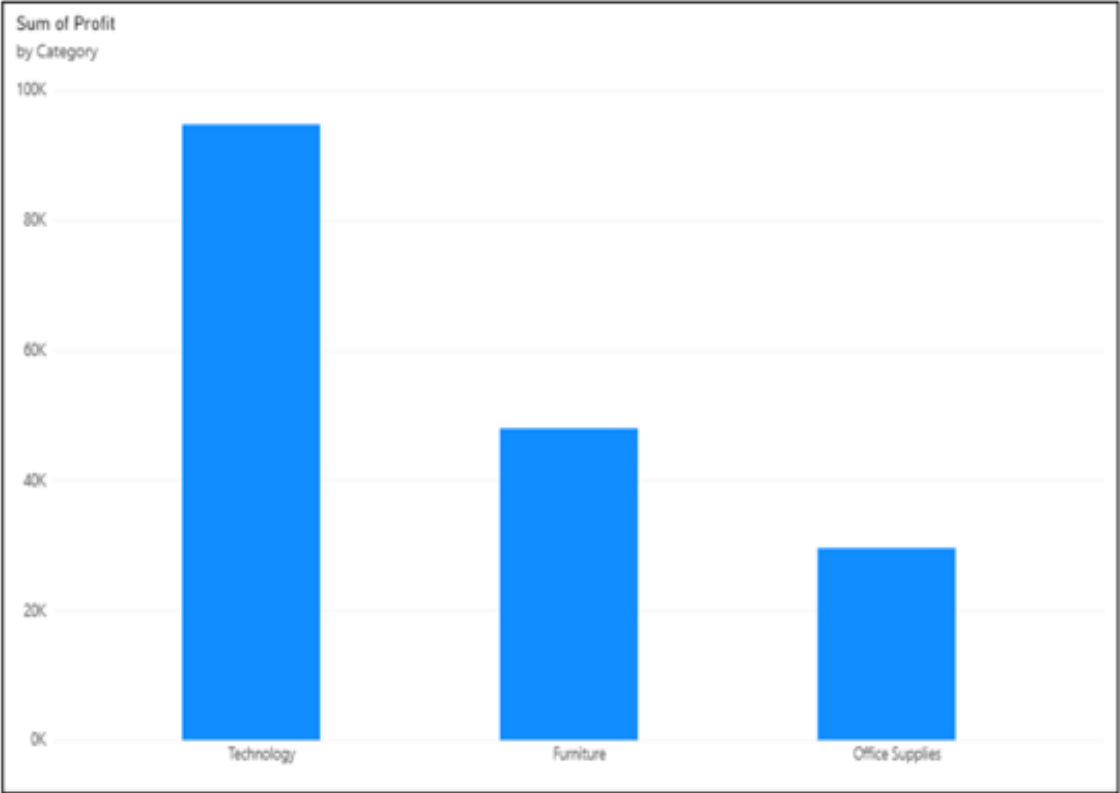
1.Sales by Region:



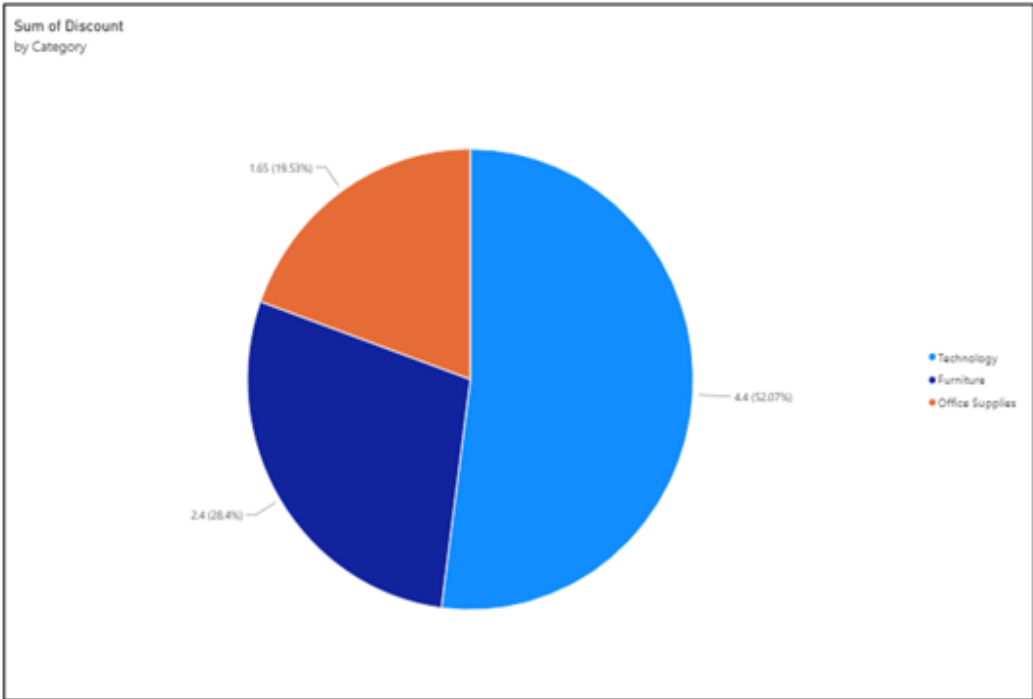
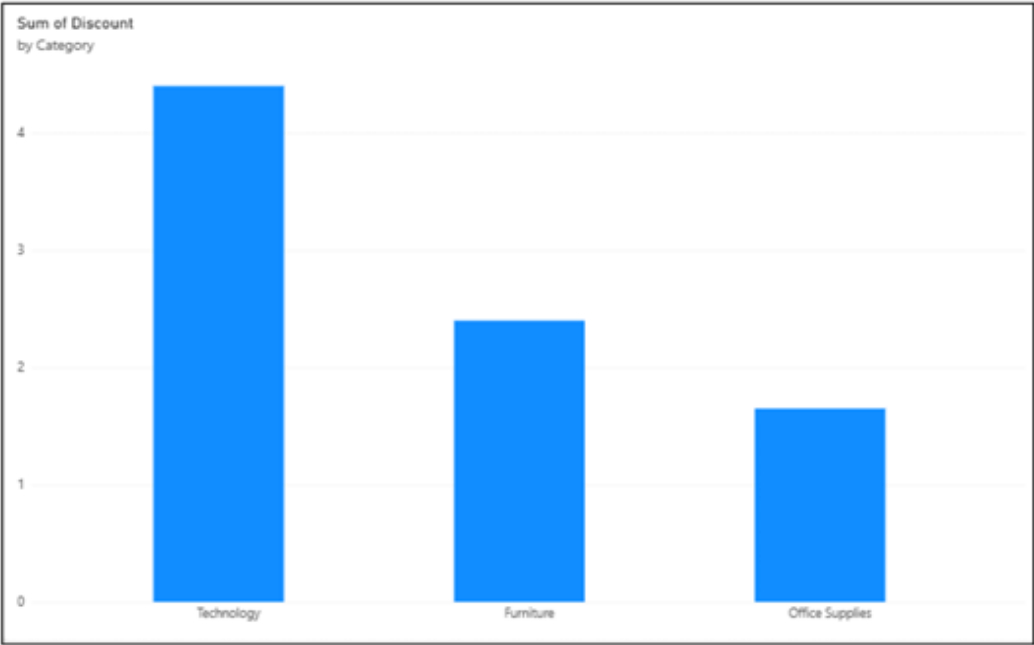


Column1	Total Sales
West	244650
South	233950
North	232350
East	223600
Total	934550

2.Profit by Category:

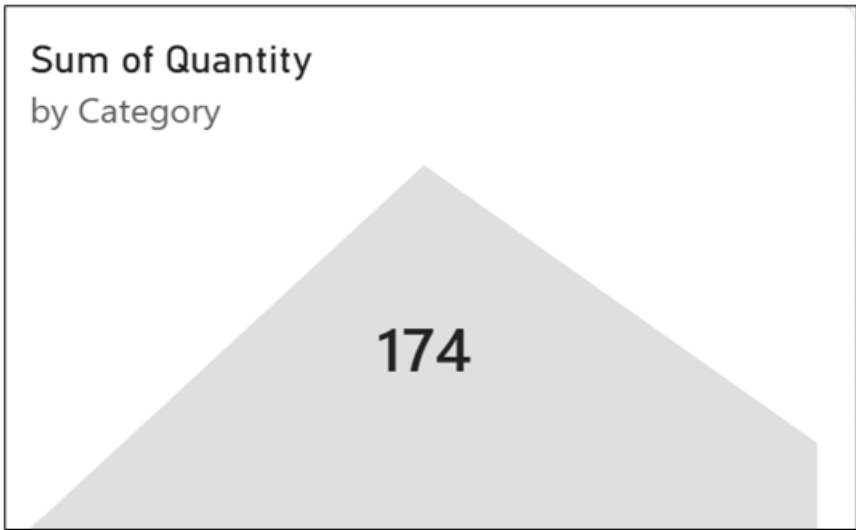
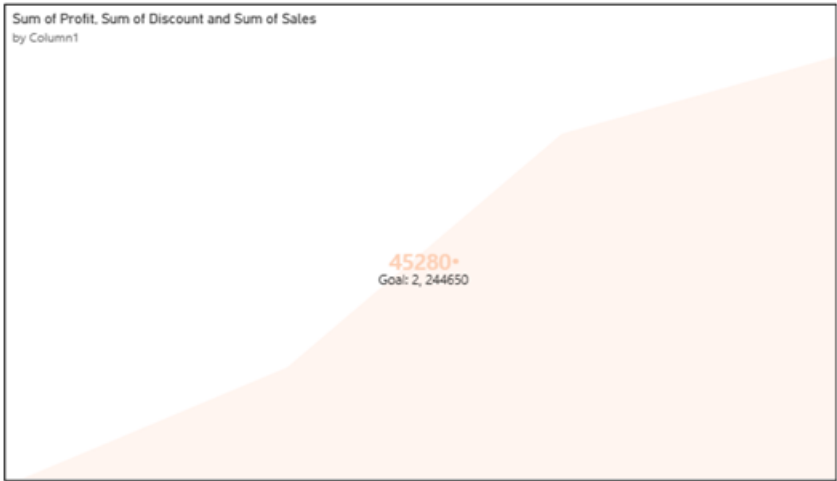
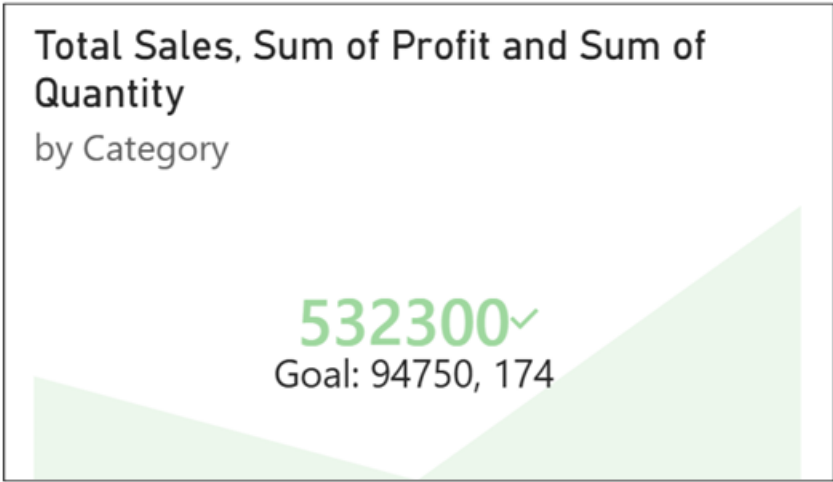


3. Discount % trend across Orders:

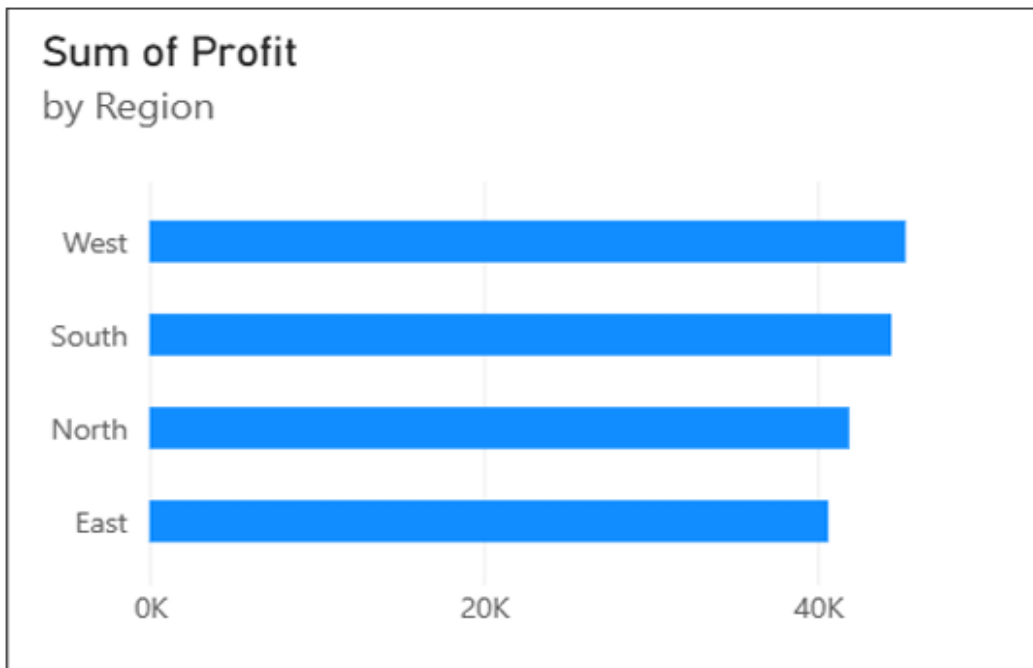


Interactive Dashboard & Filters

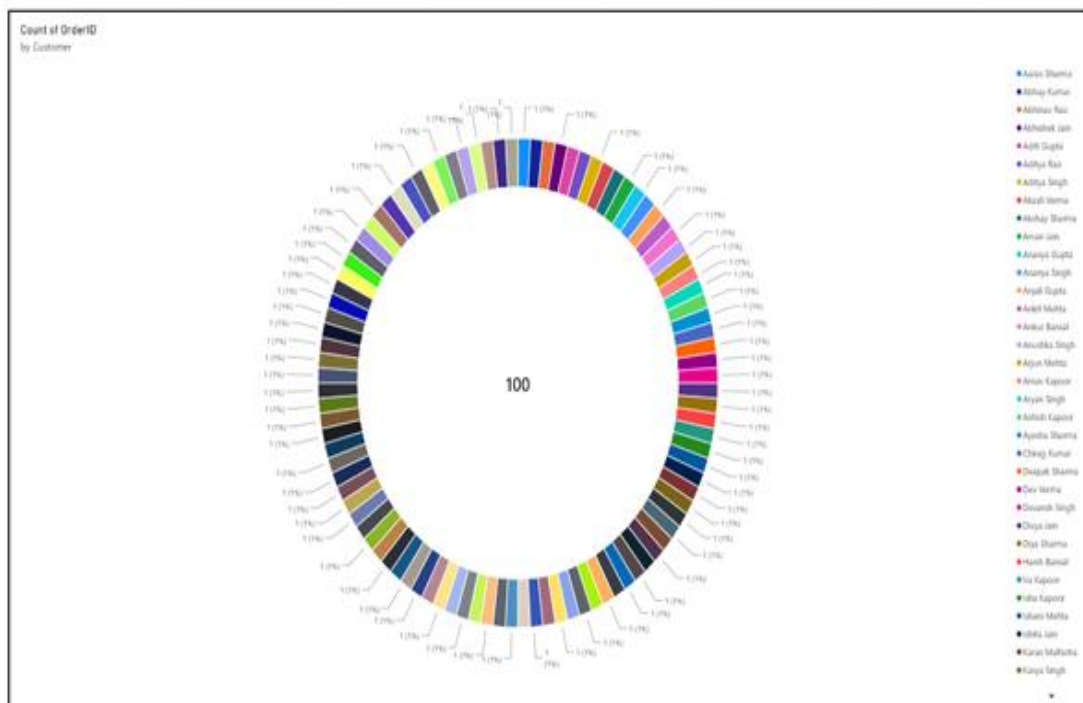
3 KPI Cards:

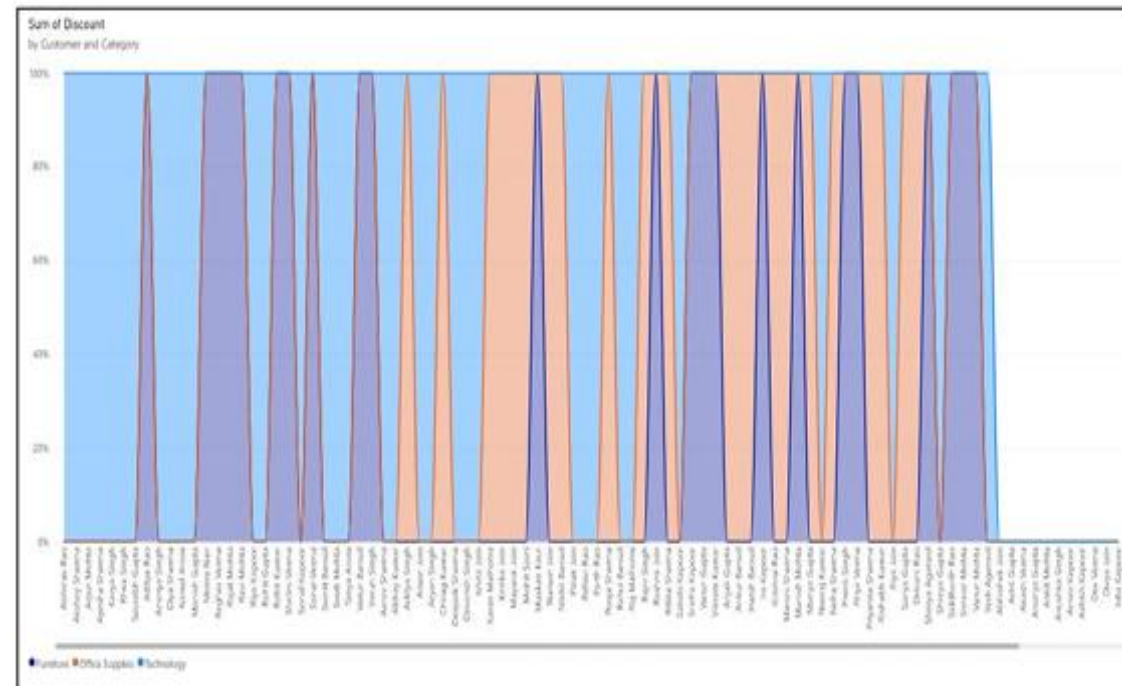


1.Stacked Bar Chart:



2. Donut Chart:





2 Slicers:

1. OrderID, Category:

OrderID

- ✓ ☐ 1001
 - > ☐ Technology
- ✓ ☐ 1002
 - > ☐ Furniture
- ✓ ☐ 1003
 - > ☐ Office Supplies
- ✓ ☐ 1004
 - > ☐ Technology
- ✓ ☐ 1005
 - > ☐ Furniture
- ✓ ☐ 1006
 - > ☐ Office Supplies
- ✓ ☐ 1007
 - > ☐ Technology
- ✓ ☐ 1008
 - > ☐ Furniture
- ✓ ☐ 1009
 - > ☐ Office Supplies
- > ☐ 1010
- > ☐ 1011
- > ☐ 1012
- > ☐ 1013
- > ☐ 1014
- > ☐ 1015
- > ☐ 1016
- > ☐ 1017
- > ☐ 1018
- > ☐ 1019
- > ☐ 1020
- > ☐ 1021
- > ☐ 1022
- > ☐ 1023

2. Category, Region:

Sum of Quantity
by Category

- ✓ ☐ Furniture
 - ☐ East
 - ☐ North
 - ☐ South
 - ☐ West
- ✓ ☐ Office Supplies
 - ☐ East
 - ☐ North
 - ☐ South
 - ☐ West
- ✓ ☐ Technology
 - ☐ East
 - ☐ North
 - ☐ South
 - ☐ West

Objective: To design an interactive dashboard in Power BI for business analysis.

Explanation: I imported the dataset into Power BI and created visuals:

- Clustered Column Chart: Sales by Region.
- Pie Chart: Profit by Category.
- Line Chart: Discount % trend across orders.
- Table/Matrix: Order details. I added KPI cards for Total Sales, Average Profit, and Profit Margin. To enhance interactivity, I inserted slicers for Region and Category, allowing dynamic filtering. The layout was organized with KPIs at the top, charts in the middle, and slicers on the side.

Outcome: A professional interactive dashboard was built, enabling users to explore sales, profit, and discount trends by region and category.

Task 7: Insights & Recommendations

Objective: To interpret the dashboard results and provide actionable recommendations for business improvement.

Explanation: After building the interactive dashboard with KPIs, charts, and slicers, I analysed the dataset to derive meaningful insights. The visuals clearly showed that the Technology category consistently outperformed others in both sales and profit. Furniture had moderate sales but weaker margins, while Office Supplies contributed the least to overall profitability.

Region-wise analysis revealed that North and East regions performed strongly, whereas South lagged behind in profit contribution. The Discount trend chart highlighted that higher discount (15–20%) directly reduced profit margins, showing a negative correlation between discounting and profitability.

Insights Derived:

1. Technology category is the most profitable and drives maximum revenue.
2. Furniture has stable sales but lower profit margins.
3. Office Supplies contribute least to profit and need improvement.
4. North and East regions are strong performers, while South requires attention.
5. High discounts reduce profit margins significantly.

Recommendations:

- Focus marketing and investment strategies on Technology to maximize revenue.

- Control discount rates to maintain healthy profit margins.
- Introduce promotional bundles or cost-cutting strategies to improve Office Supplies profitability.
- Strengthen sales efforts in the South region through targeted campaigns.

Outcome: Clear insights and practical recommendations were derived, completing the analysis cycle and showing how Power BI dashboards can guide business decisions.

Task 8: Final Report & Presentation

Objective: To compile all tasks into a structured report and present the findings professionally.

Explanation: In this final task, I consolidated the work from Task 1 to Task 7 into a single report. Each task was documented with objectives, steps performed, outcomes, and screenshots of visuals. The report highlighted the complete workflow: starting from data import and cleaning, moving into modeling, DAX calculations, visualization, interactivity, and finally insights. The dashboard screenshots were included to demonstrate the KPIs, charts, and slicers. The conclusion summarized how Power BI transformed raw data into actionable business intelligence.

Steps Performed:

- Compiled write-ups for all tasks in a structured format.
- Added screenshots of Power BI visuals (KPI cards, charts, slicers).
- Organized the report with clear headings and professional layout.
- Prepared a short presentation summary for viva/assessment.

Outcome: A complete project report was created, showcasing the entire analytical process. The final presentation demonstrated not only technical skills in Power BI but also the ability to derive insights and communicate recommendations effectively.

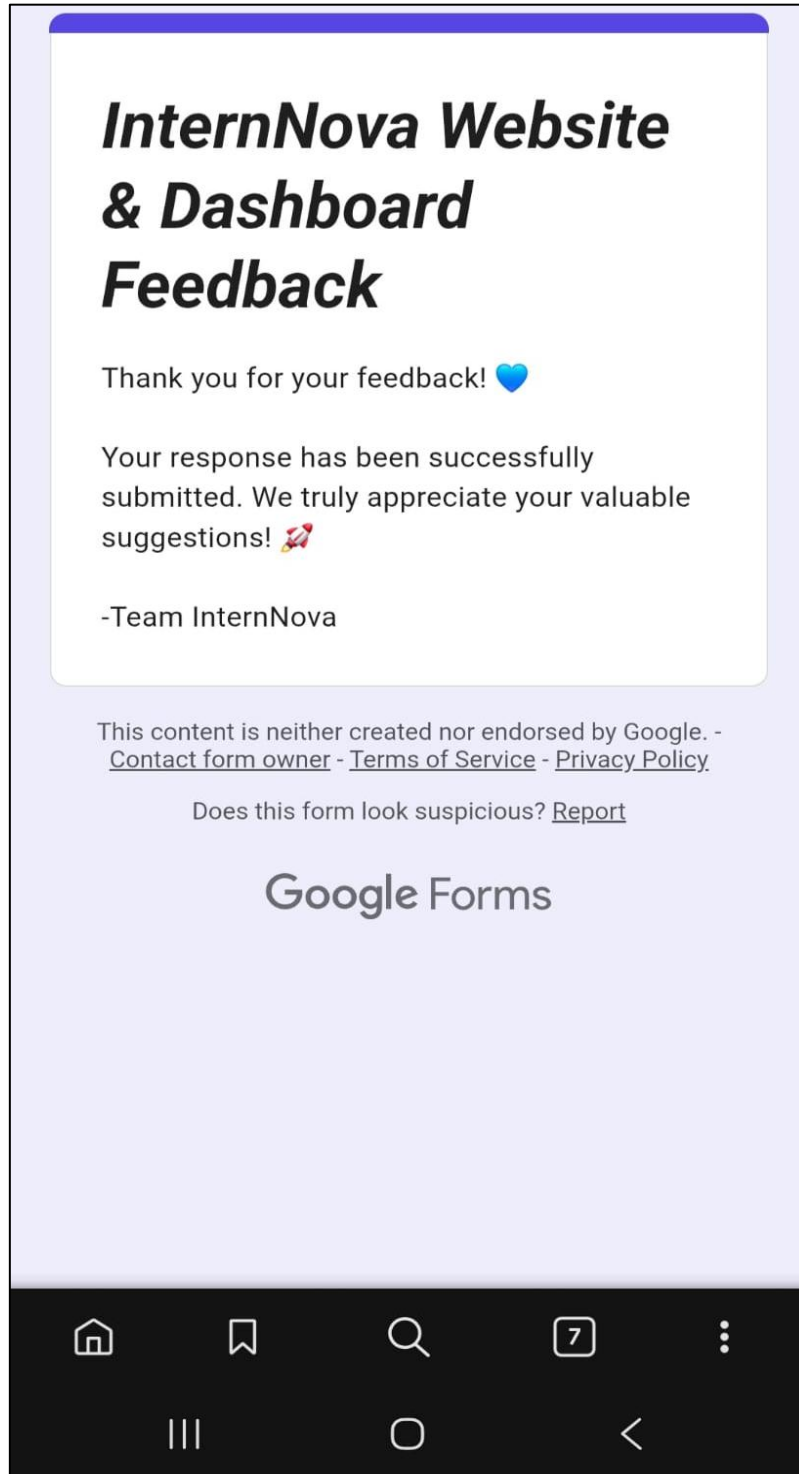
Conclusion:

Across all six tasks, I successfully demonstrated the complete data analytics workflow — from Git basics and dataset preparation to exploratory analysis, visualization, and dashboard creation. The project highlighted how combining Python, Excel, and Power BI provides a powerful toolkit for transforming raw data into actionable business intelligence. The final dashboard and insights showcase clear decision-making potential, proving the effectiveness of modern data analytics practices.

Retail Dataset Link:

https://drive.google.com/drive/folders/1yiAiuYDWXzZFaBOeVIdr_hMVba_FqbfM

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-Team InternNova

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The screenshot shows a mobile interface with a dark navigation bar at the bottom containing icons for Home, Bookmarks, Search, Tabs (7), and a Menu/More icon. Below the navigation bar are three additional icons: a vertical bar, a circle, and a left arrow.